COMP2300 Set 4 Practice Exam

Topics: ISA vs Microarchitecture, ARM/QuAC, Memory & Addressing, Control Flow, Single-Cycle Dat

Time: 120 minutes **Total:** 100 marks

Instructions

- Answer in English.
- Part A is multiple-choice (concept-focused).
- Part B is computational. Show key steps.
- Part C contains past exam questions with sources.
- Assume 32-bit ARM unless stated otherwise. Memory is byte-addressable.

Part A: Multiple Choice (30 marks, 3 marks each)

Choose exactly one option for each question.

A1. Which of the following is part of the ISA (not microarchitecture)?

(A) Ripple-carry vs carry-lookahead adder choice
(B) Instruction opcodes and semantics
(C) NAND-only vs AND/OR/NOT gate implementation
(D) Mux vs tri-state buffer choice

A2. Addressability refers to:

(A) Number of address bits
(B) Number of locations in memory
(C) Bits stored per address
(D) Total capacity in bytes

A3. In a 32-bit byte-addressable ISA, the PC increments by:

(A) 1
(B) 2
(C) 4
(D) 8

A4. Which condition flag is set when a compare result is zero?

(A) N

(B) Z
(C) C
(D) V

A5. Which statement best describes RISC?

(A) Many complex, variable-length instructions
(B) Few simple instructions, simpler hardware
(C) Optimized only for microcoded control
(D) Identical to CISC in practice

A6. Base + offset addressing computes the effective address as:

(A) Base register + (extended) offset
(B) Base register − offset
(C) PC + offset only
(D) Offset only

A7. In the ARM DP format, which bit indicates whether Src2 is an immediate?

(A) I bit (bit 25)
(B) S bit (bit 20)
(C) L bit (bit 20)
(D) Rd field (bits 15:12)

A8. The ARM branch target address (BTA) is computed as:

(A) PC + 4 + imm24
(B) PC + 8 + sign-extended (imm24 $\ll$ 2)
(C) PC + imm24 $\ll$ 1
(D) R15 + imm12

A9. For a single-cycle CPU, the clock period is determined by:

(A) The fastest instruction only
(B) The slowest instruction (critical path)
(C) The average instruction delay
(D) The number of registers

A10. The architectural state includes:

(A) Branch predictor tables only
(B) Pipeline latches only
(C) Programmer-visible registers, memory, and PC
(D) Cache replacement policy

Part B: Computational Problems (50 marks)

B1. Endianness and PC update (8 marks)
Memory bytes are shown below (addresses in hex):

Address	0x100	0x101	0x102	0x103	0x104	0x105	0x106	0x107
Byte	0x12	0x34	0x56	0x78	0x9A	0xBC	0xDE	0xF0

(a) If the system is little-endian, what 32-bit value does `LDR R0, [R1, #0]` load when $R1 = 0x100$?

(b) If the system is big-endian, what 32-bit value is loaded?

(c) If $PC = 0x100$ in a 32-bit byte-addressable ISA, what is the next sequential PC?

B2. Base + offset addressing (8 marks)
Assume the following word-addressed memory contents (each word is 32 bits):

Address	0x00	0x04	0x08	0x0C
Word	0xA1B2C3D4	0x01020304	0x11223344	0xDEADBEEF

Let $R2 = 0$.

(a) What value is loaded into $R3$ by `LDR R3, [R2, #8]`?

(b) If $R5 = 0xCAFEBABE$, what is the new value at address 0x0C after `STR R5, [R2, #12]`?

B3. Flags and conditional execution (8 marks)
Assume $R5 = 17$ and $R6 = 23$ (32-bit two's complement).

(a) After `CMP R5, R6`, determine the flags N, Z, C, V.

(b) Which of the following will execute: `SUBEQ`, `ORRMI`, `BGE`?

B4. Branch target address (8 marks)
An ARM `B` instruction is located at $PC = 0x80A0$ with `imm24`=0x000003. Compute the branch target address (BTA).

B5. Immediate extension (8 marks)
The extend block uses `ImmSrc`:

- `ImmSrc=00`: zero-extend `Instr[7:0]` (imm8)
- `ImmSrc=01`: zero-extend `Instr[11:0]` (imm12)
- `ImmSrc=10`: sign-extend `Instr[23:0]` and append `00` (imm24 $\ll$ 2)

Compute `ExtImm` for:

(a) `ImmSrc=00`, `Instr[7:0]=0xAB`

(b) `ImmSrc=01`, `Instr[11:0]=0x345`

(c) `ImmSrc=10`, `Instr[23:0]=0x00F0F0`

B6. Performance (10 marks)

(a) A program has $N = 1.2 \times 10^9$ instructions, CPI = 1.8, and clock frequency $f = 2.4$ GHz. Compute execution time.

(b) A single-cycle CPU has a clock period of 2 ns. For a program with 5×10^8 instructions, compute execution time.

Part C: Past Exam Questions (Source-Tagged) (20 marks)

C1. RISC vs CISC (Source: exam24px2.pdf, Q4, adapted) (5 marks)
Explain the difference between RISC and CISC ISAs with examples. What kind of ISA is ARM? What about QuAC?

C2. Branch prediction (Source: exam24px1.pdf, Q1h) (5 marks)
Whether or not a CPU uses branch prediction is determined by:

(A) ISA

(B) Microarchitecture

(C) Both ISA and microarchitecture

(D) Neither

C3. Execution time (Source: exam24px1.pdf, Q1j) (5 marks)
What is the execution time (in seconds) of a program that executes 2 billion instructions on a CPU with a clock frequency of 2 GHz and an average CPI of 4?

C4. Address calculation (Source: exam24px1.pdf, Q1n) (5 marks)
Register $R0$ contains the index of an element in a C array. The base address of the array is 0x0. Each element is 16 bytes. Which instruction writes the memory address of the array element into $R1$?

(A) `LSL R1, R0, #2`

(B) `LSL R1, R0, #3`

(C) `LSL R1, R0, #4`

(D) `LSL R1, R0, #5`

(E) `LSL R1, R0, #6`